

Ayomide Abilewa

Electronic and Electrical Engineering student, focused on embedded systems and instrumentation

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RESEARCH PROFILE

Most of the work so far has gone into measurement and detection under non-ideal conditions. Recent projects: a two-stage cascaded detector that fuses RGB and edge-enhanced models for pipe anomaly identification, a multi-sensor multispectral acquisition node, and control-systems practice on Quanser robotics hardware using MATLAB and Simulink.

EDUCATION

B.Sc. Electronic and Electrical Engineering · Obafemi Awolowo University 2021–2027 (expected)
Ile-Ife, Osun State, Nigeria

Relevant coursework: Control Systems, Measurement and Instrumentation, Digital Electronics, Signal Processing, Power Systems, Microprocessors and Embedded Systems, Electronic Circuit Design.

RESEARCH INTERESTS

- Detection under degraded conditions: How to keep detection precision steady when lighting, surface condition and viewing angle vary within a single run, instead of tuning for one condition and losing the others.
- Measurement and calibration: Turning raw sensor output into a measurement that means something: calibration, separating quantities that are easy to conflate, and the signal standards industry already relies on.
- Feedback and control: Closed-loop behaviour on real hardware, from a thermistor-driven PWM loop to PID control on Quanser platforms, where the actuator changes the quantity being measured.
- Embedded and real-time vision: Running vision pipelines within real constraints: modest hardware, live video, and the requirement that the system respond now rather than after a batch job.
- Systems that degrade rather than fail: Designing explicit fallback behaviour so losing a network, a model or a sensor costs quality instead of function.

RESEARCH AND TECHNICAL PROJECTS

Live Video Pipe Anomaly Detection · Python, YOLOv8, YOLOv11, Weighted Boxes Fusion 2026–Present

- Developing a two-stage cascaded detection architecture for real-time pipe anomaly identification, targeting industrial inspection.
- Assembled a parallel ensemble of RGB and edge-enhanced YOLO models merged with Weighted Boxes Fusion to improve precision across varied lighting and surface conditions.

Multispectral Acquisition Device for Hydroponics · ESP32, GY-2561, GY-31, I2C 2025

- Built a multi-sensor ESP32 acquisition node measuring light intensity and colour spectrum with the GY-2561 and GY-31, streaming live to a Blynk IoT Cloud dashboard.
- Integrated multiple I2C sensors on a single ESP32, handling bus arbitration and sensor calibration.

AI-Powered Smart Attendance System · Python, OpenCV, YuNet, SFace Feb–Mar 2026

- Owned the computer-vision pipeline and Flask REST API backend for an AI-powered attendance system using YuNet face detection and SFace recognition with cosine similarity matching.
- Evaluated recognition performance under real classroom lighting and off-axis faces rather than curated frontal images.
- Defended the system at a final-year group project review and was invited to develop an extended production version.

github.com/ayomide-abilewa/Smart-Attendance-System

Smart Temperature-Controlled Fan · Arduino, C++, PWM, NTC thermistor

2024

- Implemented continuous proportional actuation rather than threshold switching to avoid overshoot.

RESEARCH AND ENGINEERING EXPERIENCE

Research and Development Intern (SIWES) · ACE Quanser Robotics Lab, Obafemi Awolowo University
Ile-Ife, Nigeria

- Operated the QDrone quadrotor UAV, QCar autonomous ground vehicle and QArm 6-DOF manipulator using MATLAB and Simulink.
- Applied PID control, sensor feedback and real-time system analysis to live hardware rather than simulation alone.
- Delivered a technical orientation presentation on the Quanser Aero 2 covering system architecture, control principles and laboratory safety.

Facilities Engineering Management Intern (SIWES) · Chevron Nigeria Limited

Mar 2025–Present

Lagos, Nigeria

- Rotate across the Electrical and Instrumentation (E&I) team, working on oil and gas field instrumentation and electrical systems.
- Interpret P&IDs and study control loop configurations, pneumatic signal standards (3–15 PSI), I/P converters and hazardous area classification.

Embedded Systems and EEE Tutor · Astar Tutorials

2023–Present

Ile-Ife, Nigeria

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, JavaScript, Verilog/HDL, Bash

Embedded and Hardware: Arduino, ESP32, STM32, Raspberry Pi, FPGA, I2C / SPI / UART, Sensor integration

AI and Machine Learning: YOLOv8/v11, OpenCV, TensorFlow (fundamentals), PyTorch (fundamentals),
Weighted Boxes Fusion

Instrumentation and Control: P&ID interpretation, PID control loops, Pneumatic systems, I/P converters, 4–20
mA and 3–15 PSI standards

Tools: VS Code, Git/GitHub, MATLAB/Simulink, KiCad, Flask, React/TypeScript, PostgreSQL, Blynk IoT

TRAINING AND CERTIFICATIONS

FPGA Design and HDL/Verilog Workshop (SWEP) · Faculty of Engineering, Obafemi Awolowo University

2024

Robotics Lab Orientation and Operations · ACE Quanser Lab, Obafemi Awolowo University

2024

TEACHING AND ACADEMIC SERVICE

SPAW Project Lead and Syllabus Team Lead · IEEE OAU Student Branch

2024–Present

- Lead SPAW 3.0, a six-session embedded systems and entrepreneurship curriculum for secondary school students.